The Microscopic System Behind Bacri *et al.* (1995)

Morphology, Magnetism, and Field Response of a Cobalt-Ferrite Ferrofluid

Scope. This report reconstructs the *physical* ferrofluid in Bacri, Perzynski, Shliomis & Burde (*Phys. Rev. Lett.* **75**, 2128, 1995): how the colloid is built, what one particle looks like and does, what "the particles align" means mechanically, and how fields reshape the magnetic and structural state. Sources: the three project papers (Bacri 1995; Shliomis–Morozov, *Phys. Fluids* **6**, 2855, 1994; Shliomis–Stepanov, *JMMM* **122**, 176, 1993) plus the colloid/magnetism literature. All numbers are computed for Bacri's own particles. Critical caveats are boxed.

1 What the colloid is, physically

Bacri's fluid is a **concentrated aqueous ionic ferrofluid of cobalt-ferrite nanoparticles**, made by Massart's method (Neveu, Jussieu). Three facts fix it: material $\mathbf{CoFe_2O_4}$ with grain magnetization $M_s = 400$ G and anisotropy $K = 2 \times 10^6$ erg/cm^3; diameter $d \approx 10$ nm; volume fraction $\varphi = 20\%$ in water. Each choice is deliberate.

1.1 How it is made

1. **Coprecipitation.** Fe^{2+}/Fe^{3+} (plus Co^{2+}) salts precipitated by strong base at high pH nucleate spinel-ferrite nanocrystals directly from solution.
2. **Surface treatment / peptization.** Nitric-acid (and ferric-nitrate) treatment protonates the oxide surface to $\equiv$Fe–OH_2^+, giving each grain a surface charge (~ 0.2 C/m^2).
3. **Electrostatic stabilization.** Charged grains repel—no surfactant needed ("ionic" ferrofluid). Bacri's specific Co-ferrite/citrate variant is citrate-coated, negatively charged at neutral pH.

The payoff: ionic ferrite sols reach $\varphi \sim 0.26$ while staying stable and Newtonian—why Bacri could run $\varphi = 0.20$ to maximize the signal.

1.2 What one particle looks like

A grain is a roughly spherical single-crystal spinel-ferrite nanoparticle carrying a *permanent* moment, wrapped in a thin charged shell (Fig. 1). Its moment, for Bacri's particle:

$$m = M_s V = M_s \tfrac{\pi}{6} d^3 \approx 2.1 \times 10^{-16} \text{ emu} \approx 2.3 \times 10^4 \, \mu_B. \tag{1}$$

One grain is a single "macrospin" of ~20,000 Bohr magnetons—the object every equation treats as "the particle."

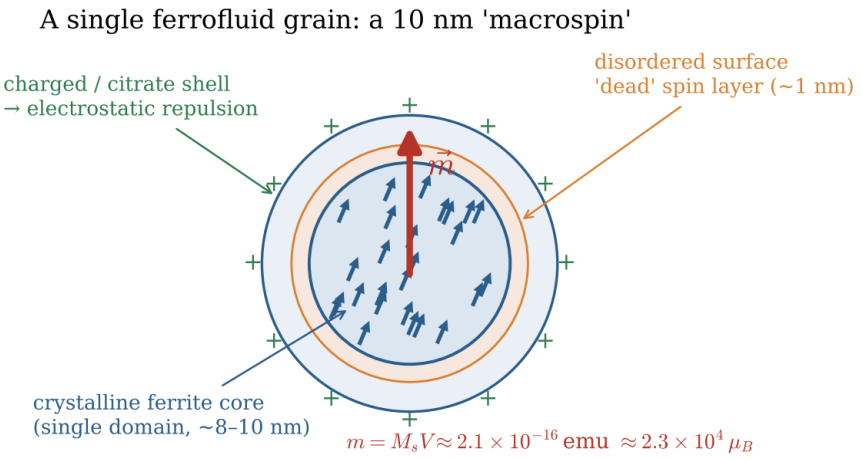


Figure 1: A single grain. Thousands of exchange-locked atomic moments (blue) sum to one net classical vector $\vec{m}$ (red). The ferrite core is single-domain; a ∼1 nm disordered "dead" layer lowers the effective M_s; the outer charged/citrate shell provides the electrostatic repulsion that keeps the colloid stable and sets the closest approach between particles.

1.3 Polydispersity

Real coprecipitated colloids are not monodisperse; size follows a Gamma distribution $P(d) = d^\beta/[d_0^{\beta+1}\Gamma(\beta+1)]\, e^{-d/d_0}$, $\bar{d} = (\beta+1)d_0$, with $\beta \sim 2.5$–15 (Shliomis–Stepanov). Because $m \propto d^3$ and relaxation times depend *exponentially* on d^3, a modest width in d spreads relaxation times over many decades.

2 Why 10 nm, single-domain, superparamagnetic—and why cobalt ferrite

2.1 Single-domain

Below a critical diameter $d_{SD} \sim \sqrt{A/K}$, a domain wall costs more than uniform magnetization, so the grain is single-domain: one uniform magnetization, no walls. For Co-ferrite the single-domain window extends to ∼40 nm; at 10 nm Bacri's particles are safely single-domain.

2.2 The anisotropy barrier and the Néel time

A single-domain moment prefers an easy axis fixed in the crystal; tilting it by θ costs $E_a = KV\sin^2\theta$, a barrier KV. Compare to kT via $\sigma \equiv KV/kT$. The thermal hop time is the **Néel time**

$$\tau_N = \tau_0\, e^{\sigma} = \tau_0\, e^{KV/kT}, \qquad \tau_0 \sim 10^{-9}\,\text{s}. \tag{2}$$

For Bacri's particle ($K = 2\times10^6$, $V = 5.2\times10^{-19}$ cm^3, $kT = 4.1\times10^{-14}$ erg):

$$\sigma = KV/kT \approx 25.3, \qquad \tau_N \approx 10^{-9} e^{25.3} \approx 10^2\ \text{s}. \tag{3}$$

The Néel time is ∼100 seconds. The moment is utterly frozen to the crystal—the paper's "rigid dipole behavior."

Why Co-ferrite, not magnetite. At the same 10 nm, magnetite ($K \approx 1.35 \times 10^5$) gives $\sigma \approx 1.7$, $\tau_N \approx 5.5$ ns—the moment flips internally in nanoseconds and the particle body need not move. That would *kill* the magnetoviscous effect. **Only rigid dipoles transmit magnetic torque into mechanical torque on the fluid.** Co-ferrite's large K raises σ to $\sim$25, freezing the moment and forcing the body to carry it. That is the entire point of the material choice.

2.3 Two relaxation channels, one winner

A moment realigns by either route, acting in parallel ($\tau^{-1} = \tau_N^{-1} + \tau_B^{-1}$):

Mechanism	What moves	Time
Néel	moment rotates *inside* a stationary crystal	$\tau_N = \tau_0 e^{KV/kT}$
Brownian	*whole particle* rotates, moment fixed to it	$\tau_B = 3\eta V_h/kT$

For Co-ferrite $\tau_N \sim 100$ s $\gg \tau_B$, so Brownian rotation wins decisively (Fig. 2). Bacri *measured* $\tau_B = 1.6$ ms by transient birefringence—larger than the $\sim$1 µs a water estimate gives, reflecting the high local viscosity of the dense sol. Either way the hierarchy holds: **Brownian tumbling of the physical particle is the only magnetization-relaxation channel that operates.**

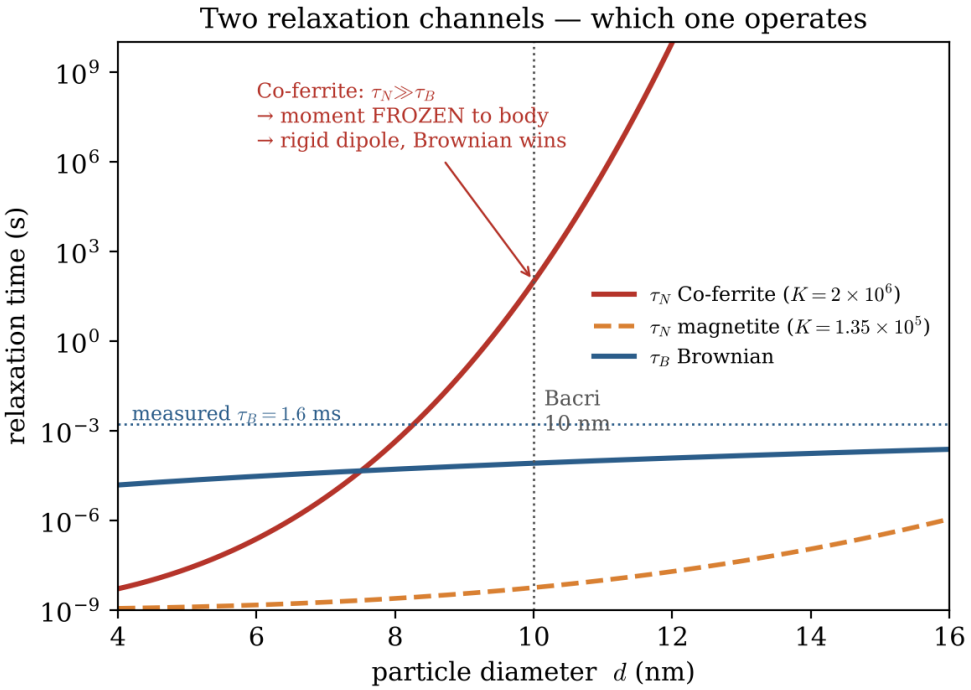


Figure 2: The two relaxation times vs. particle size. Néel time (red/orange) depends *exponentially* on d^3 and on material K; Brownian time (blue) depends weakly on size. For Co-ferrite at 10 nm, τ_N sits $\sim$5 orders above τ_B: the moment is frozen, and the particle relaxes only by physically rotating. For magnetite the curves cross near 10 nm, so its moment would relax internally instead.

For your LBM: `compute_M` uses a single τ_B as *the* relaxation time. This is legitimate *only* because Co-ferrite is in the frozen-Néel limit. Change material or size and the Néel channel reopens; the single-τ_B magnetization equation then silently becomes wrong.

3 What "the particles align" actually means

The phrase covers three physically distinct things. Keeping them separate is essential.

3.1 Moment alignment (orientational, single-particle): Langevin

In zero field, thermal tumbling randomizes the giant moments; net magnetization is zero despite each particle being permanently magnetic (ensemble superparamagnetism). A field exerts an aligning torque opposed by thermal agitation, with competition set by the **Langevin parameter** $\xi = mH/kT$ and equilibrium alignment

$$\frac{M}{M_{\text{sat}}} = L(\xi) = \coth \xi - \frac{1}{\xi}. \tag{4}$$

"The particles align" primarily means this distribution tilts toward H (Fig. 3). At Bacri's 2000 Oe, $\xi \approx 10.1$, so $L \approx 0.9$: moments ∼90% aligned.

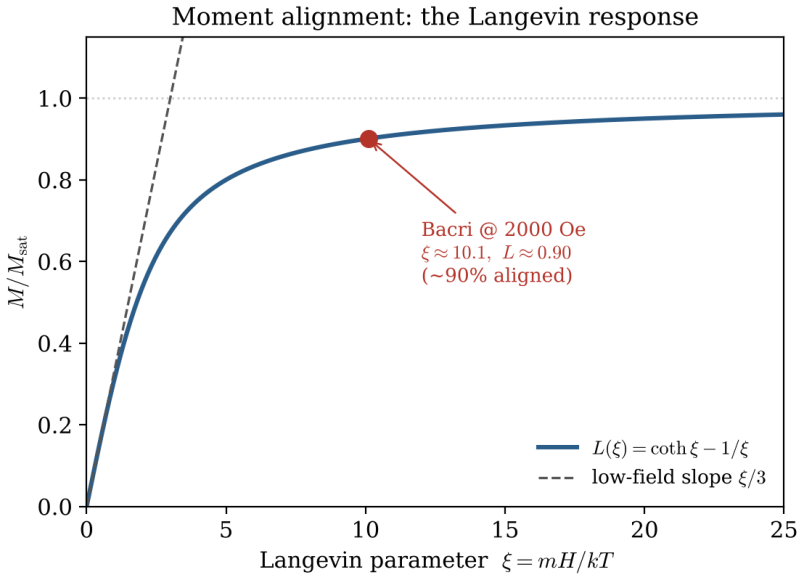


Figure 3: Langevin magnetization. Low field: linear ($M \propto \xi/3$). High field: saturates. Bacri at 2000 Oe sits deep in saturation ($\xi \approx 10$).

Rigid-dipole subtlety. Because the moment is frozen to the body, aligning the *moment* means physically **rotating the whole grain** so its frozen easy axis points along H. Moment alignment and body orientation are the *same event*—which is exactly why the field can mechanically interfere with the particle's rotation in a flow (§4).

3.2 Structural alignment (positional, many-particle): chains

Separately, particles attract via dipole–dipole coupling and may form head-to-tail chains. The governing ratio is the **dipolar coupling constant** $\lambda = m^2/(d^3 kT)$; $\lambda \gtrsim 1$ allows chains, $\lambda \gg 1$ gives strong chaining. Computed for Bacri (Fig. 4): $\lambda \approx 1.06$ at bare contact, falling to 0.6–0.8 once the ∼1–2 nm shell is included.

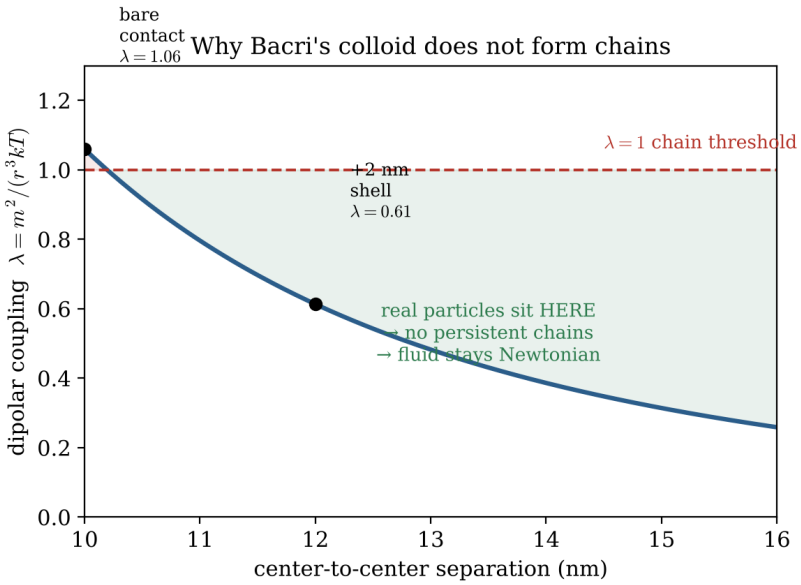


Figure 4: Dipolar coupling vs. center-to-center separation for Bacri's particles. The stabilizing shell holds centers $\gtrsim$11–12 nm apart, pushing the *effective* λ below the chaining threshold. The colloid is at most marginally chain-forming.

The central structural point. Bacri's particles sit at $\lambda \approx 1$ at bare contact and *below* 1 once the shell prevents contact. So the colloid forms at most transient pairs—no long columns. The companion Co-ferrite/citrate system (same lab) reports "essentially no chain structures." The theory invoked (Shliomis–Morozov) is explicitly *dilute single-particle* ("neglect both hydrodynamic and magnetic interparticle interactions"), yet $\varphi = 20\%$. Reconciliation: *magnetically* dilute ($\lambda \lesssim 1 \Rightarrow$ weak dipolar correlations, moments respond independently) even while *hydrodynamically* dense (zero-field η is already 77× the solvent). The theory handles the latter by using measured η, φ, τ_B as inputs.

Consequence for your reproduction. Your single-particle closure matching Bacri to ∼0.3% (static)/∼1.6% (AC) is itself *evidence* that interparticle magnetic correlations are negligible—an independent confirmation of the no-chains morphology. If chains mattered, a single-particle model could not match. State this explicitly in your defense.

4 How the external field reshapes the system

4.1 The torque balance (the heart of it)

In shear, a fluid element spins at $\vec{\Omega} = \frac{1}{2}\nabla \times \vec{v}$. A free particle co-rotates ($\vec{\omega}_p = \vec{\Omega}$, no slip, no extra dissipation). The field breaks this. Bacri Eq. (1):

$$6\eta\varphi\,(\vec{\omega}_p - \vec{\Omega}) = \vec{M} \times \vec{H}. \tag{5}$$

Under a **static field** the moment (hence body) is held along H, resisting tumbling: the particle lags the fluid, which must shear past it ⇒ extra dissipation, **rotational viscosity** $\Delta\eta > 0$ (Fig. 5), saturating as

$$\Delta\eta = \tfrac{3}{2}\eta\varphi\,\frac{\xi - \tanh\xi}{\xi + \tanh\xi}\sin^2\beta \;\xrightarrow{\xi\to\infty}\; \tfrac{3}{2}\eta\varphi, \tag{6}$$

with β the angle between H and $\vec{\Omega}$ (maximal at $\beta = \pi/2$). Bacri sees 77→220 cP under 2000 Oe DC. A permanent field can only *increase* viscosity—it does no work, so it can only impede.

$$6\eta\varphi\,(\vec{\omega}_p - \vec{\Omega}) = \vec{M} \times \vec{H}$$

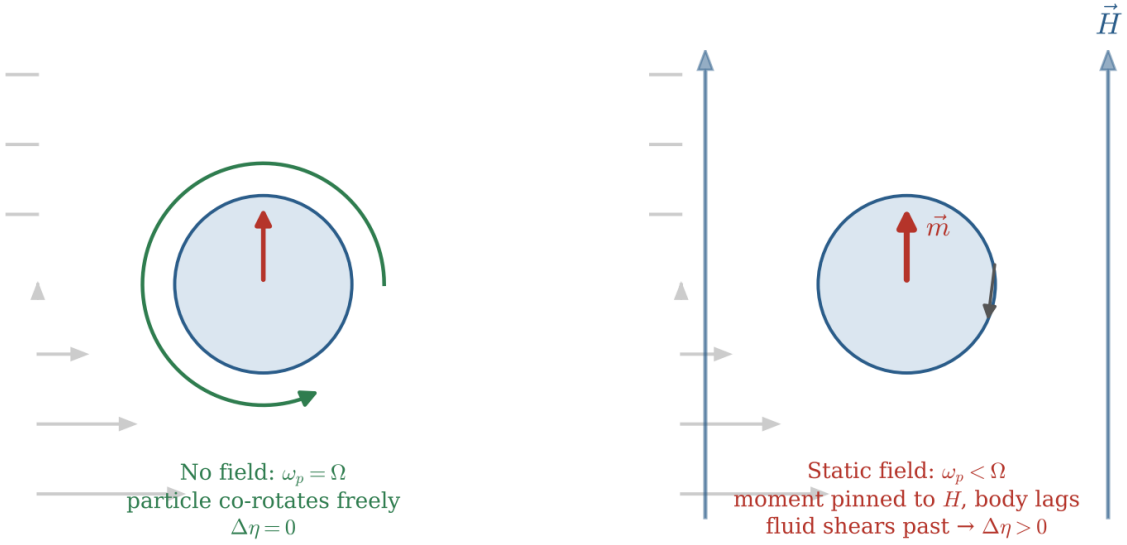


Figure 5: Torque balance in a shear flow (grey = velocity gradient). **Left:** no field—the grain co-rotates with the local vorticity, zero extra drag. **Right:** a static field pins the frozen moment along $\vec{H}$; the grain can no longer keep up with the flow's rotation, so the fluid shears past it and dissipates extra energy ($\Delta\eta > 0$).

4.2 The AC twist: why viscosity goes negative

Under $H(t) = H_0 \cos \omega t$: in quiescent fluid the field drives rotational *swings* with no preferred sense—decompose into two counter-rotating circular fields that cancel, so $\langle \omega_p \rangle = 0$. **Any vorticity breaks the symmetry**: the co-rotating component grips better, a net $\langle \omega_p \rangle \neq 0$ appears. The sign depends on frequency:

- **Slow** ($\omega\tau_B < 1$): moment tracks the field, behaves quasi-statically $\Rightarrow \Delta\eta > 0$.
- **Fast** ($\omega\tau_B > 1$): the moment lags so far that over a cycle the field drives the grain to spin *faster* than the fluid ($\omega_p > \Omega$). The grain now pumps angular momentum *into* the flow. AC-field energy → particle spin → fluid kinetic energy: $\Delta\eta < 0$.

Sign change at $\omega_0\tau_B \approx 1$; minimum at $\omega_m\tau_B = \sqrt{3} \approx 1.73$ where $\Delta\eta = -\frac{1}{8}\Delta\eta(0)$ (Fig. 6). Bacri tunes total viscosity 220 → 50 cP purely by AC frequency.

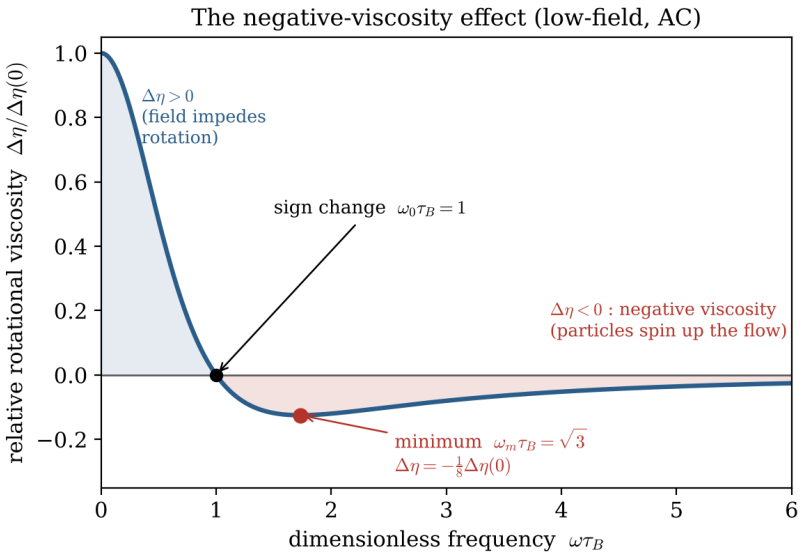


Figure 6: The negative-viscosity effect (low-field, Shliomis–Morozov). $\Delta\eta$ changes sign at $\omega_0\tau_B = 1$ and reaches its deepest negative value at $\omega_m\tau_B = \sqrt{3}$. In the red region the alternating field is feeding energy into the flow, lowering the measured viscosity below its zero-field value.

4.3 Magnetization dynamics: why simple relaxation isn't enough

Debye/Shliomis relaxation (Bacri Eq. 4),

$$\frac{d\vec{M}}{dt} = \vec{\omega} \times \vec{M} - \frac{1}{\tau_B}(\vec{M} - \vec{M}_0) - \frac{1}{6\eta\varphi}\vec{M} \times (\vec{M} \times \vec{H}), \quad \vec{M}_0 = \varphi M_s L(\xi)\hat{h},$$

works for DC or slow/weak AC. For finite ξ *and* finite $\omega\tau_B$—Bacri's regime—one needs the **effective-field (MRSh) equation** (Bacri Eqs. 5–6): the actual nonequilibrium $\vec{M}$ is taken equal to the equilibrium magnetization evaluated at an effective field ξ_e obeying its own ODE. This is what `compute_M` implements, and what reproduces the non-monotonic field dependence and the correct negative-viscosity minimum.

Known validity limits (already in your notes): PRL Eq. (3) has a misprint; S–M Eq. (40) has a missing square; and the 2D MRSh closure gives the *wrong sign* of g at strong AC drive ($\xi_0 = 10$, $\omega\tau_B = 1.5$)—a real limitation of truncating the effective-field expansion at strong, fast drive, not a code bug. Cite these as boundaries of validity.

4.4 Structural response (for completeness)

In strongly dipolar fluids ($\lambda \gg 1$) a field also drives chain formation and lateral coalescence into columns—a morphological transition with large, often non-Newtonian rheology. **Bacri's fluid does not do this** (§3.2): it stays Newtonian with and without field. The whole magnetoviscous response here is orientational/rotational and single-particle; the Newtonian behavior under field is itself the experimental signature that chains are absent.

5 The microscopic state at a glance

Quantity	Symbol	Value	Meaning
Material	$CoFe_2O_4$	—	high-K ferrite → rigid dipole
Diameter	d	10 nm	single-domain, blocked
Grain magnetization	M_s	400 G	core moment density
Moment / particle	$m = M_s V$	2.1×10^{-16} emu ($2.3 \times 10^4 \mu_B$)	the "macrospin"
Anisotropy ratio	$\sigma = KV/kT$	≈ 25	$\gg 1$: moment frozen to body
Néel time	τ_N	≈ 100 s	internal flips impossible
Brownian time	τ_B	1.6 ms (measured)	only active relaxation channel
Langevin @2000 Oe	ξ	≈ 10	moments $\sim$90% aligned
Dipolar coupling	λ	≈ 1 bare, < 1 coated	marginal: essentially no chains
Volume fraction	φ	0.20	hydro. dense, magn. dilute
Sign-change freq.	$\omega_0 \tau_B$	≈ 1	onset of negative viscosity
Min-viscosity freq.	$\omega_m \tau_B$	$\sqrt{3} \approx 1.73$	deepest negative $\Delta\eta$

6 The one-paragraph physical story

Bacri's ferrofluid is a dense (20 vol%) water suspension of 10 nm cobalt-ferrite crystals, each a single magnetic domain carrying a permanent $\sim$20,000-μ_B moment, kept from clumping by a thin charged shell. Cobalt ferrite's large anisotropy freezes each moment rigidly to its crystal (Néel time $\sim$100 s), so the particle is a *rigid dipole*: to turn the moment you must turn the whole grain, and the moment relaxes only by the grain tumbling (Brownian, $\sim$1.6 ms). In zero field the grains tumble randomly—no net magnetism, no preferred spin. A field aligns the moments (and hence the grains); in shear this fights the flow's tendency to tumble them, so a steady field makes grains lag the fluid and raises viscosity. An AC field drives swings with no preferred sense until a flow's vorticity breaks the tie; at high frequency the lagging moment ends up spinning the grains *faster* than the fluid, injecting the field's energy back into the flow and dropping the measured viscosity even below its zero-field value. Because the particles are only marginally dipolar-coupled ($\lambda \lesssim 1$) they never build chains, the fluid stays Newtonian, and the whole effect is cleanly a single-particle rotational phenomenon—which is why a single-particle theory, and your single-particle LBM closure, can reproduce it.

Primary sources. Bacri, Perzynski, Shliomis, Burde, *Phys. Rev. Lett.* **75**, 2128 (1995). Shliomis & Morozov, *Phys. Fluids* **6**, 2855 (1994). Shliomis & Stepanov, *JMMM* **122**, 176 (1993). Massart, *IEEE Trans. Magn.* **17**, 1247 (1981). Rosensweig, *Ferrohydrodynamics* (CUP, 1985). Supporting λ/morphology data: Neveu-lab Co-ferrite/citrate characterization; dipolar-chain simulation & scattering literature (Ivanov–Kantorovich; Klokkenburg *et al.*).